MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

III SEMESTER B.TECH. (INFORMATION TECHNOLOGY/ COMPUTER AND COMMUNICATION ENGINEERING) END SEMESTER EXAMINATIONS, March, 2021

SUBJECT: OBJECT ORIENTED PROGRAMMING [ICT 2155]

REVISED CREDIT SYSTEM
(1/ 03 /2021)

Time: 3 Hours MAX. MARKS: 50

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data if any, may be suitable assumed.

1A.	Create a swing UI to accept a number from the user and on click of the button: i. First verify if the user has entered a valid number, if not display a message to the user asking him to input a valid number. ii. If the number is valid display square of the number and also state if the number is prime or not. The message should be displayed in a label. The contents of the label is empty initially.	**5**
1B.	Create a class Vehicle, derive classes Ship & Plane from class Vehicle. In main method create objects of Vehicle, Ship & Plane and when the display method is called it should display "moves on land" for Vehicle object. "moves in air" for plane object & "moves on water" for ship object. The following conditions should be strictly considered while defining the classes: i. The display method is present only in the Vehicle class ii. No usage of constructors in any of the class	**3**

1C.	Write the output for the given code snippets:
	
	i.
	`public class Test`
	`{`
	`public static void leftshift(int i, int j)`
	`{`
	`i <<= j;`
	`}`
	`public static void main(String args[])`
	`{`
	`int i = 4, j = 2;`
	`leftshift(i, j);`
	`System.out.println(i);`
	`} }`
		2

	ii. class SC2 {
	public static void main(String [] args) {
	SC2 s = new SC2();
	s.start(); }
	
	void start() {
	int a = 3; int b = 4;
	System.out.print(" " + 7 + 2 + " ");
	System.out.print(a + b);
	System.out.print(" " + a + b + " ");
	System.out.print(foo() + a + b + " ");
	System.out.println(a + b + foo()); }
	String foo() { return "foo"; }}	

2A	Write a java program to do the following: i. Accept a word from the user, create two threads ii. In the first thread, arrange the characters of the word alphabetically & display the word iii. In the second thread, count the no. of vowels in the word and display it iv. Once the two threads are complete, display " WORK COMPLETED"	**5**
2B	Create a list of 5 book objects. Each book object is associated with a Title, Author, Price and ISBN. Write a complete Java program to arrange the 5 books in alphabetical order of their title and display them (display all fields of each book).	**3**

2C	Write the output for the following code snippet with proper justification:
	
	i. public class ex_ha {
	static String str= "a" ;
	public static void main(String args[]){
	new ex_ha().method1();
	System.out.println(str); }
	void method1() {
	try {
	method2();
	} catch(Exception e) {
	str += "b"; }
	finally { str +="u"; } }
	
	void method2() throws Exception{
	try { method3();
	str += "c";
	}catch(Exception e) {
	throw new Exception();
	}finally { str +="d"; }
	method3();
	str += "e";
	}
	void method3() throws Exception
	{ throw new Exception();
	} }
		2

	ii.
	i. class MM{
	static {System.out.println('A');}
	public MM(){
	System.out.println('C');
	}
	}
	
	class NN extends MM{
	static {System.out.println('D');}
	public NN(){
	System.out.println('F');
	}
	}
	public class teststatic {
	public static void main(String[] args){
	NN n = new NN();
	}
	}	

3A.	Write a java program to read the file "word.txt" containing set of words. Check each word, whether it contains vowels or not. If the word contains vowels, copy that word into the file "vowels.txt" otherwise copy it into "nonvowels.txt". Also display the number of lines in the file "word.txt".	**5**
3B.	Create an interface called "TelecomServices" with methods "voicecharges" and "datacharges". Create classes "Prepaid" and "Postpaid" that use the interface "TelecomServices". Calculate the voice charges and data charges for both Prepaid and Postpaid connections using appropriate methods. The methods should use the variables "callrate" and "duration" in order to find out the service charges.	**3**
3C.	State the benefits of Swing over AWT. Write a note on two types of containers defined by swing.	**2**
4A.	Create a class COVIDReport with sample id, symptom and test_result as its members. The test_result will be either "CORONA POSITIVE" or "CORONA NEGATIVE". Create a class Patient having patient id, name, age and an object of COVIDReport as its members. Patient class also has a method displayReport, which displays the patient details with a prescription message. If the patent age is more than 50 and test_result is "CORONA POSITIVE", then prescription will be "HOSPITALIZED" else "HOME QUARANTINED." Use appropriate constructors.	**5**
4B.	Consider two integer arrays **Troop1[ ]** and **Troop2[ ].** Each array consisting of **N** integer values represent the marching speed of two troops travelling in the same direction, at each time unit. Write a program to find the maximum total distance travelled by the two troops together(side by side) throughout the journey. Example: **Input:** Troop1[ ] = {11,12, 13, 12, 14}, Troop2[ ] = {12, 11, 13, 11, 14} **Expected output:** 36	**3**

4C.	What is dynamic method dispatch? How does it differ from compile time polymorphism?	**2**
5A.	Write a Java program to read a character matrix of dimension greater than 3 x 3 and an word (String). In the entered matrix find if the given word is present in horizontal forward, horizontal backward, vertical forward and vertical backward direction. Example: Input to the program: B A T C T T O P S E X Y A W A A E W Z X If search word = TOP **Expected Ouput**: "TRUE" Horizontal Forward If search word = TEA **Expected Ouput**: "TRUE" Vertical Forward If search word = WAY **Expected Ouput**: "TRUE" Horizontal Backward If search word = APT **Expected Ouput**: "TRUE" Vertical Backward And so on…	**5**
5B.	What is method overloading and method overriding? Explain with a suitable example	**3**
5C.	Two String objects with identical values are not equal when == operator is used. Explain.	**2**